Response to the Editorial Return

exdqlm: An R Package for Estimation and Analysis of Flexible Dynamic Quantile Linear Models

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Dear Editors,

Thank you for the careful second prescreening review. We understand the return as a reproducibility-interface and output-provenance problem: the previous archive still made it too hard to run the article computations in batch mode and compare the resulting output directly with the manuscript. We have restructured the replication materials accordingly and are resubmitting as a new submission with a point-by-point response.

The submitted files are:

- `exdqlm-jss.pdf`: manuscript PDF.
- `exdqlm_1.1.1.tar.gz`: software source.
- `exdqlm-jss-replication.tar.gz`: replication archive.
- `response-to-editor.pdf`: this response.

The package source is version 1.1.1 of exdqlm. This is a narrow reproducibility and inference-stability patch: it does not change the statistical model, exported API, or manuscript claims. It corrects compiled stochastic helper paths so they use serial R-controlled random-number streams, adds tests for repeated seed behavior, and sets more stable default exAL scale–skewness updates for MCMC and LDVB.

Summary of the Revision

- We replaced the previous replication interface with one flat reviewer-facing `code.R`. The script does not call `source()`, read or write saved fit objects, inspect Git metadata, or require local paths.
- The primary command is the JSS-style batch command `R CMD BATCH -vanilla code.R code.Rout`, with native thread variables set before R starts. The run produces `code.Rout` and `Rplots.pdf` at the archive root.
- The full `code.Rout` prints the fitted object `M95`, `summary(M95)`, `MTF$median.kt`, Tables 7–10, and `sessionInfo()`. This makes the numerical output visible in the batch output instead of

only in separate files.

- The graphics stream produced by `code.R` contains the manuscript figures in manuscript order. The same plotting expressions also write the PNG files used by the manuscript.
- The manuscript now includes a computational-details paragraph describing the reference R version, exdqlm version, RNG settings, backend/thread settings, and the role of the full replication script.
- The README and reproducibility notes now describe the public replication command and expected outputs. Development-only scripts remain in the repository for author maintenance but are not included as reviewer-facing entry points.

Responses to the Specific Comments

Comment 1. Single script and batch-mode output

The revised archive centers on one R script, `code.R`. It is a flat, commented file ordered by the manuscript examples. Running `R CMD BATCH -vanilla code.R code.Rout` performs the full manuscript replication and creates the batch output file requested in the editorial comment.

We removed the ambiguous `examples.R` entry point from the JSS-facing archive. The submitted archive now gives reviewers one obvious script to run and one batch output file to inspect.

Comment 2. Exact reproducibility and computing environment

The public script sets

```
RNGversion("4.6.0")
```

and

```
RNGkind("Mersenne-Twister", "Inversion", "Rejection").
```

It also stops immediately unless the loaded package is exdqlm version 1.1.1.

While investigating the differences reported by the editors, we found that some compiled stochastic helper routines in version 1.1.0 could use OpenMP worker threads or private C++ seeds in ways that were not appropriate for exact seed-based replication. Version 1.1.1 corrects these paths by using serial R-controlled random-number streams. The same patch also uses an exact scale-collapsed gamma update for exAL MCMC and a structured $q(\gamma)q(\sigma \mid \gamma)$ scale–skewness factor for LDVB. We added package tests for repeated-seed behavior in the compiled samplers, dynamic MCMC, static MCMC, and the structured scale–skewness factor, and the package passes `R CMD check -as-cran` under R 4.6.0.

The manuscript and README now state the reference environment and the thread variables used before launching R. Runtime values are still described as platform-dependent elapsed fitting times.

Comment 3. More output in the manuscript and batch output

We added visible fitted-object output to the replication script and manuscript discussion. The full batch output now includes

```
M95
```

and

```
summary(M95).
```

The script also prints Tables 7–10 and `MTF$median.kt` directly to `code.Rout`. This allows a reviewer to inspect the generated batch output without searching through intermediate files.

Comment 4. Figure 1 and Figure 9 differences

Both figures are now regenerated by `code.R` from the same plotting functions used to write the manuscript PNG files. The package patch in version 1.1.1 removes the compiled stochastic RNG/thread behavior that could make MCMC-derived outputs differ under repeated seeds or different thread settings. The response does not claim this was the only possible source of the previously observed visual differences; instead, the revised archive makes the figure provenance direct and rerunnable from the submitted script.

Comment 5. `MTF$median.kt`

The manuscript value is now taken from the full `code.R` reference run. The scalar is printed explicitly in `code.Rout`, so it can be compared directly with the manuscript.

Comment 6. Missing Figure 3 top panel and Figure 5

`code.R` now regenerates the full graphics stream in manuscript order. The stream includes the complete Sunspots figure with its top panel and the Big Tree data/covariate figure. The corresponding PNG files are written under `figures/`.

Comment 7. Missing Tables 7 and 10

Tables 7 and 10 are now generated by the full script and printed to `code.Rout`. The same table objects are also saved as CSV files under `tables/`.

Comment 8. `tab.ex3` and `tab.ex3.fc`

The Big Tree fitted-diagnostic table and held-out forecast table are now printed to `code.Rout` as Tables 8 and 9. They are also saved as CSV files in the generated table directory.

Comment 9. Fitted-object output

The full script prints the `M95` object and `summary(M95)`. The manuscript now guides the reader through representative fitted-object output, so the standard `print()` and `summary()` methods are visible in the article and in the batch output.

Validation

Before resubmission we ran the following checks with R 4.6.0:

- targeted package repeatability tests for compiled samplers, dynamic MCMC, and static MCMC;
- the package test suite and CRAN submission checks for `exdqlm_1.1.1.tar.gz`;
- `R CMD BATCH -vanilla code.R code.Rout`;
- manuscript and response compilation; and
- archive extraction and execution checks from a directory without Git metadata.

We believe the revised materials now provide a simpler and more standard JSS replication interface, with the numerical and graphical outputs visible in the batch files requested by the editors.

Sincerely,

Antonio De Leon, Raquel Barata, Raquel Prado, and Bruno Sansó